Department of Computer Science and Engineering (CSE)
BRAC University

# Practice Problem Set 6

## CSE251 – Electronic Devices and Circuits

## BJT CIRCUITS

Logic Function Implementation, Modes of Operation, Method of Assumed States, Multistage Circuits, MOSFET-BJT Hybrid Problems

*$I_B$, $I_C$, and $I_E$ denote the base, collector, and emitter currents, respectively. Similarly, $V_B$, $V_C$, and $V_E$ denote the corresponding voltages. "Sat" and "Active" refer to the saturation and active regions, respectively. $k_n$ and $k_n'$ denote MOSFET's transconductance and process transconductance parameters, respectively. $V_{T_n}$ denote the threshold voltage of the MOSFETs. $\beta$ denote the common emitter current gain of BJTs.*

Course Description, COs, and Policies

BRAC UNIVERSITY
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Midterm and Final Questions

For any error or issues, contact purbayan.das@bracu.ac.bd

# Problem 1

- Give a switch-BJT implementation of the following logic functions. $A$, $B$, $C$, $D$, $E$ and $F$ are the Boolean inputs and $f$ is the Boolean output. Here, "+" and "." represent the logical $OR$ and $AND$ operations, respectively.

I. $f = A.\bar{B}$

II. $f = \overline{A.B.C + D.E}$

III. $f = \overline{A + B.C + D}$ [Summer 2024]

IV. $f = A.B + \bar{A}.\bar{B}$ [Fall 2024]

V. $f = \overline{A.C} + \overline{B + C}$

VI. $f = (A.B + C).D$

VII. $f = A.B + B.C + C.A$

VIII. $f = (A + B).(C + D)$

IX. $f = (A + B).C + D.E$

X. $f = \bar{A}.B + \bar{B}.A + \bar{C}.A$ [Spring 2025]

XI. $f = (A.B + C).D.(E + F)$

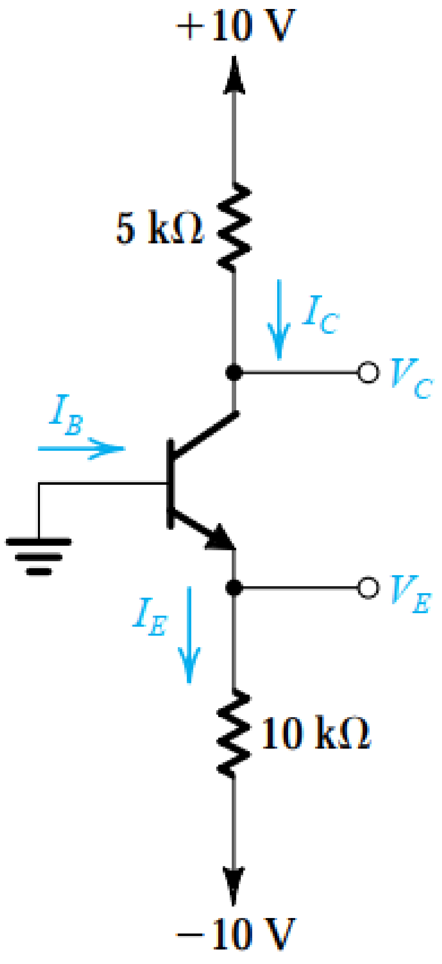
XII. $f = A \oplus B$

XIII. $f = \overline{(A + B + C).D + \bar{D}}$

XIV. $f = \overline{C.(A + B)}.(A + \bar{B} + C)$

# Problem 2

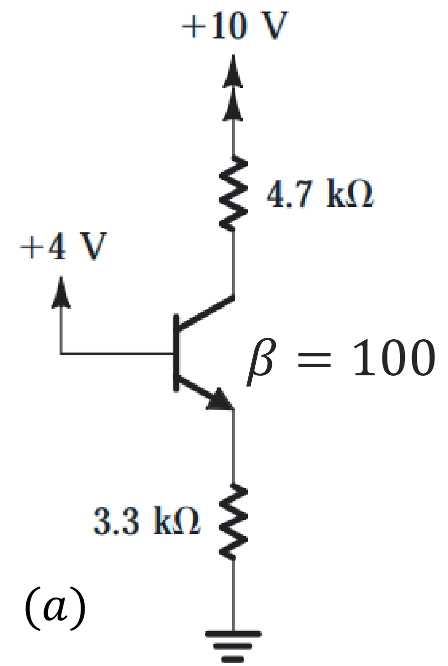
- For the adjacent circuit, $V_E$ was measured and found to be $-0.7\ V$. If $\beta = 50$, determine $I_E$, $I_B$, $I_C$, and $V_C$.

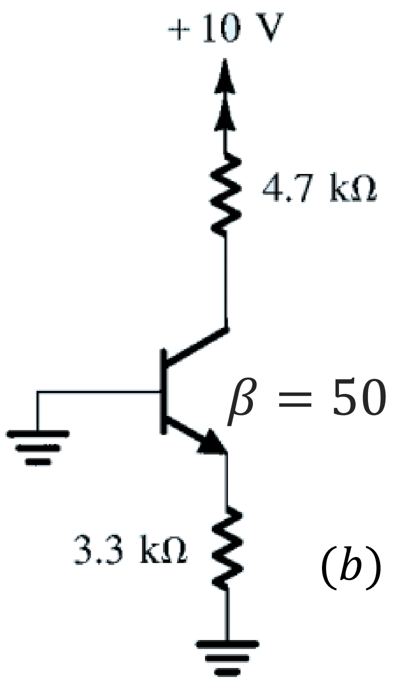


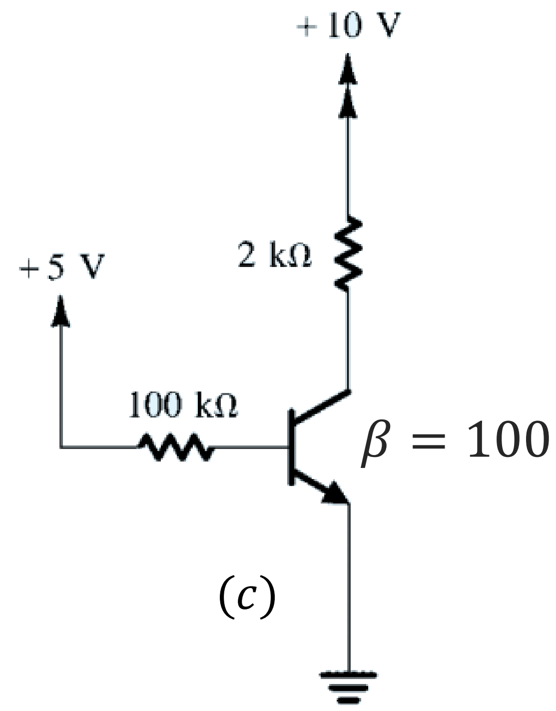

Ans: $\boldsymbol{Active; I_E = 0.93\ mA; I_B = 18\ \mu A; I_C = 0.91\ mA; V_C = 5.45\ V; V_E = -0.7\ V}$

# Problem 3

- Determine and verify the operating region for each of the transistors in the following circuits. Assume $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.8\,V$, and $V_{CE,\,sat} = 0.2\,V$.



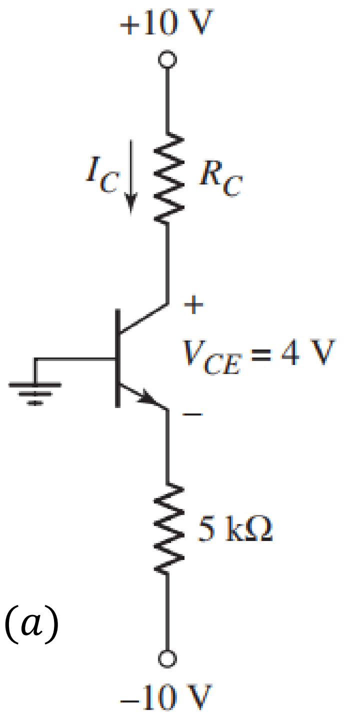

Ans: $\boldsymbol{Active; I_C = 0.99\,mA; I_E = 1\,mA; I_B = 10\,\mu A; V_B = 0\,V;\ V_C = 5.347\,V; V_E = 3.3\,V}$

Ans: $\boldsymbol{Cutoff; I_C = I_E = I_B = 0; V_B = V_E = 0\,V;\ V_C = 10\,V}$

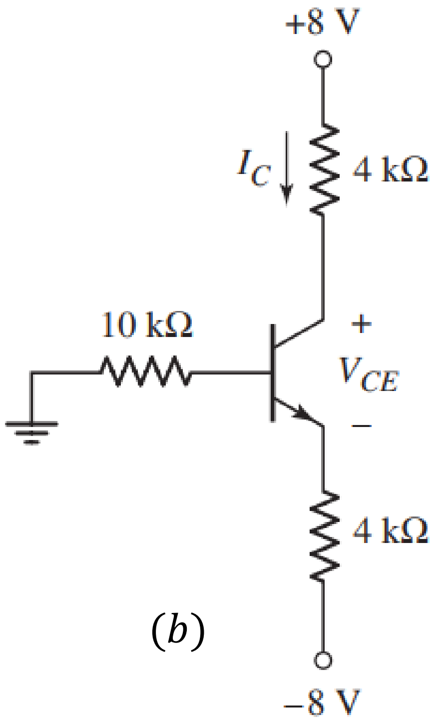
Ans: $\boldsymbol{Active; I_C = 4.3\,mA; I_E = 4.343\,mA; I_B = 43\,\mu A; V_B = 0.7\,V;\ V_C = 1.4\,V; V_E = 0\,V}$

# Problem 4

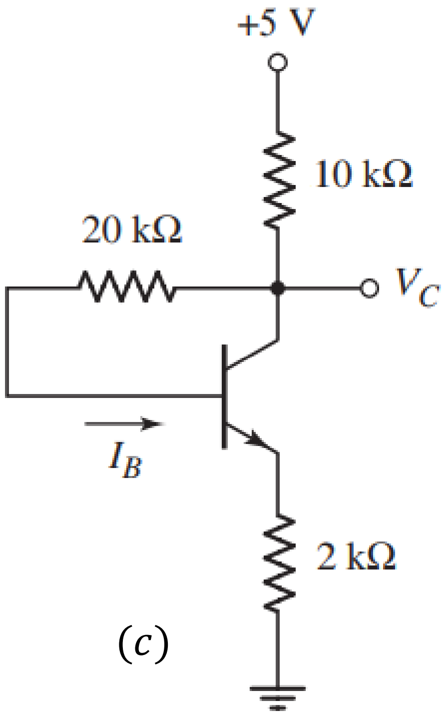
- For the NPN transistors in the following circuit, $\beta = 75$, $V_{BE,\,active} = 0.7\ V$, $V_{BE,\,sat} = 0.8\ V$, and $V_{CE,\,sat} = 0.2\ V$. Determine the labeled quantities.




$(a)$




$(b)$




$(c)$

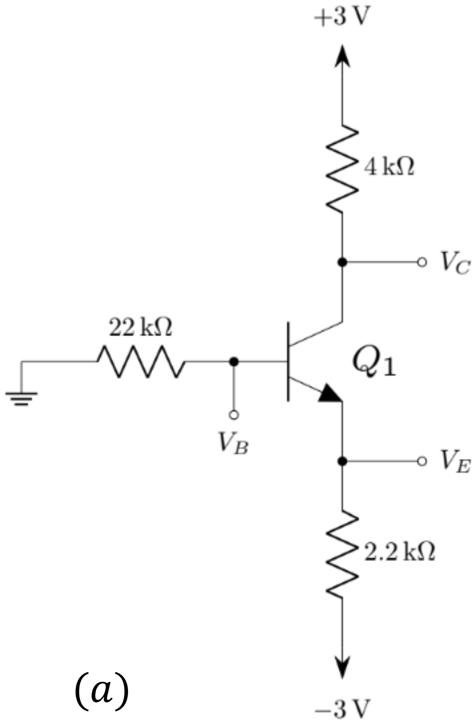
Ans: $\boldsymbol{Active; I_C = 1.84\ mA; I_E = 1.86\ mA; I_B = 20\ \mu A; V_B = 0\ V;\ V_C = 3.3\ V; V_E = -0.7\ V; R_C = 3.64\ k\Omega}$

Ans: $\boldsymbol{Active; I_C = 1.725\ mA; I_E = 1.76\ mA; I_B = 35\ \mu A; V_B = -0.26\ V;\ V_C = 1.1\ V; V_E = -0.96\ V}$

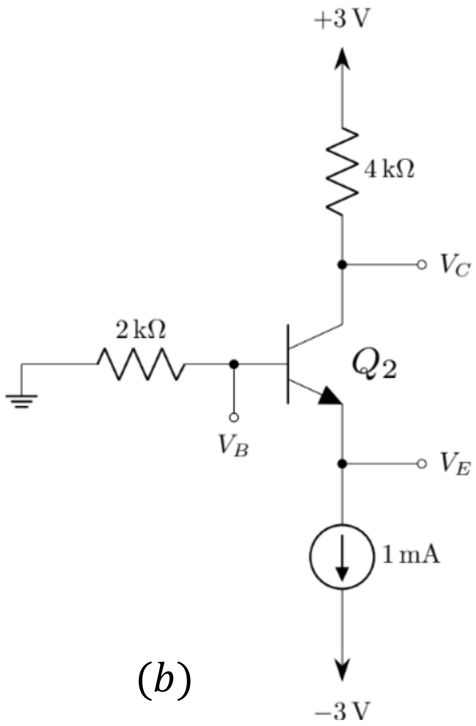
Ans: $\boldsymbol{Active; I_C = 0.341\ mA; I_E = 0.346\ mA; I_B = 5\ \mu A; V_B = 1.4\ V;\ V_C = 1.54\ V; V_E = 0.7\ V}$

# Problem 5

- For the transistors in the following circuits, $\beta = 100$, $V_{BE,\,active} = 0.7\ V$, $V_{BE,\,sat} = 0.8\ V$, and $V_{CE,\,sat} = 0.2\ V$. Determine $V_B$. $V_C$, and $V_E$. *[Hint for (b): $V_{BC} = V_{BE} - V_{CE}$]*




$(a)$



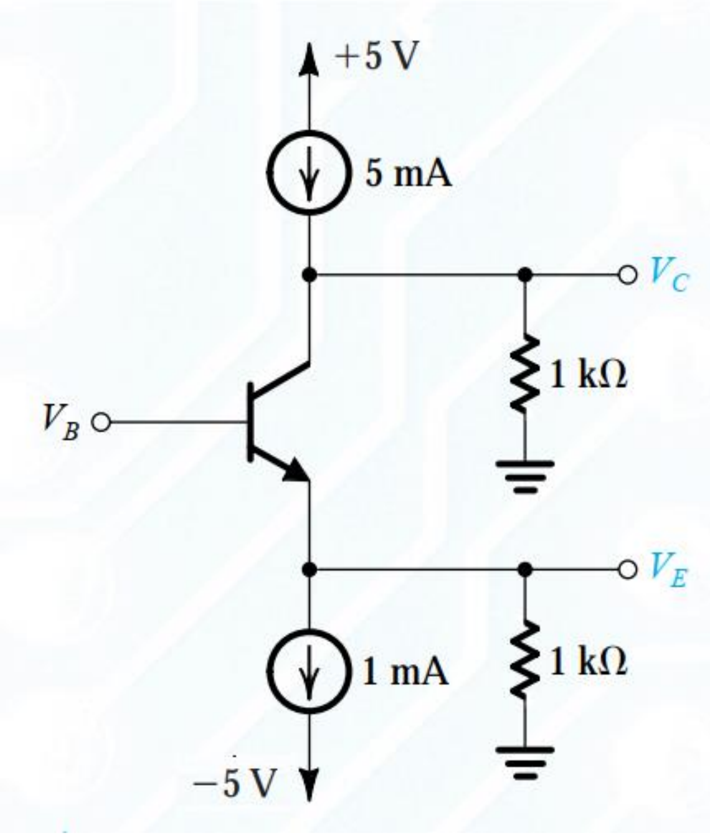

$(b)$

Ans: $\boldsymbol{Sat}$; $\boldsymbol{I_C = 0.933\ mA}$; $\boldsymbol{I_E = 0.9394\ mA}$; $\boldsymbol{I_B = 6.1\ \mu A}$; $\boldsymbol{V_B = -0.13\ V}$; $\boldsymbol{V_C = -0.732\ V}$; $\boldsymbol{V_E = -0.93\ V}$

Ans: $\boldsymbol{Sat}$; $\boldsymbol{I_C = 0.933\ mA}$; $\boldsymbol{I_E = 1\ mA}$; $\boldsymbol{I_B = 0.067\ mA}$; $\boldsymbol{V_B = -0.134\ V}$; $\boldsymbol{V_C = -0.732\ V}$; $\boldsymbol{V_E = -0.934\ V}$

# Problem 6

- For the adjacent circuit, assume that $V_{BE} = 0.7\,V$. For what value of $V_B$ does the transistor operates in cut-off?




Ans: $\boldsymbol{I_C = I_B = I_E = 0;\ V_B = 0.3\,V;\ V_C = 5\,V; V_E = -1\,V}$

# Problem 7

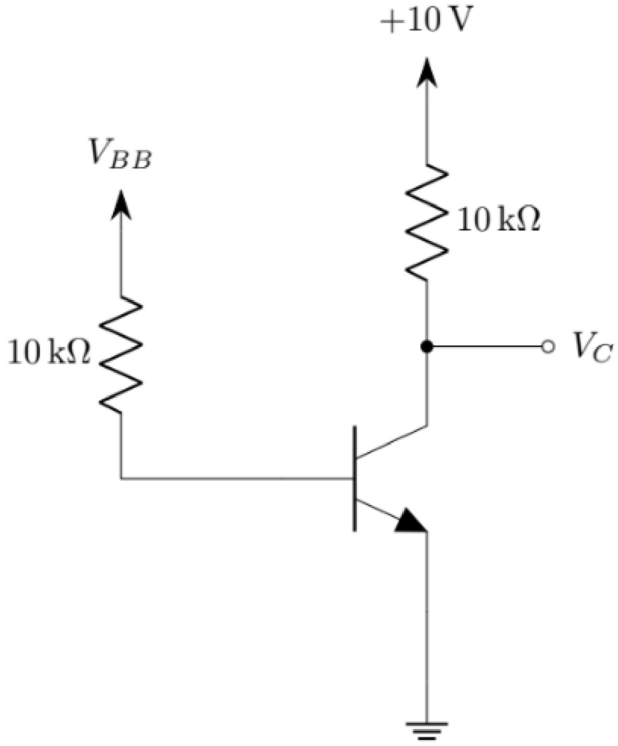
- For the adjacent circuit, assume that $V_{BE}$ remains constant at $0.7\,V$ when the transistor is ON. The transistor $\beta$ is specified to be 50. Determine the value of $V_{BB}$ that results in the transistor operating:

  I. in the active region with $V_{CE} = 5\,V$,

  II. at the edge of saturation, and

  *[Hint: at the edge of saturation, $\beta$ is still 50 and assume $V_{CE} = 0.3\,V$.]*

  III. deep in saturation with $\beta_{forced} = 10$ and $V_{CE} = 0.2\,V$.




Ans: $I.\, I_C = 0.5\,mA; I_E = 0.51\,mA; I_B = 10\,\mu A; V_B = 0.7\,V;\ V_C = 5\,V; V_E = 0\,V; V_{BB} = 0.8\,V$

Ans: $II.\, I_C = 0.97\,mA; I_E = 0.9894\,mA; I_B = 19.4\,\mu A; V_B = 0.7\,V;\ V_C = 0.3\,V; V_E = 0\,V; V_{BB} = 0.894\,V$

Ans: $III.\, I_C = 0.98\,mA; I_E = 1.076\,mA; I_B = 98\,\mu A; V_B = 0.7\,V;\ V_C = 0.2\,V; V_E = 0\,V; V_{BB} = 1.68\,V$

# Problem 8

- For the following circuit, $V_S = 5\ V$, $R_L = 10\ k\Omega$, and $R_I = 500\ k\Omega$. Assume $\beta = 100$ for the transistor, $V_{BE,\,active} = V_{BE,\,sat} = 0.6\ V$, and $V_{CE,\,at\ the\ edge\ of\ sat} = V_{CE,\,sat} = 0.2\ V$.

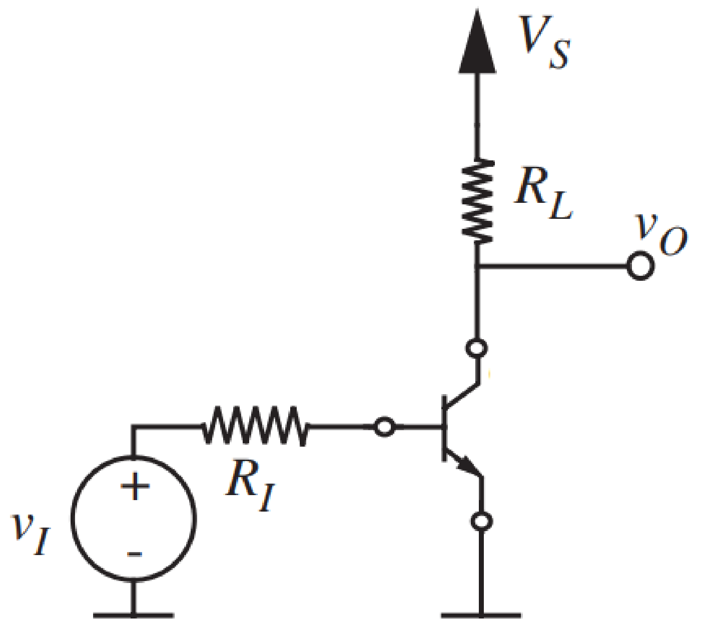
I. Derive an expression relating $v_O$ and $v_I$ assuming the transistor is in active mode. Ans: $\boldsymbol{v_O = 6.2 - 2v_I}$

II. What is the highest value of $v_I$ for which the transistor operates in cut-off. Ans: $\boldsymbol{0.6\ V}$

III. What is the highest (or lowest) value of $v_I$ for which the transistor operates in its active region (or enters into its saturation region). What are the corresponding range for $v_O$? *[$\beta = 100$ at the edge of saturation]* Ans: $\boldsymbol{3\ V}$

IV. Based on the relation found in I and values of $v_I$ and $v_O$ found in II and III, draw the VTC ($v_O\ vs.\ v_I$) graph for the circuit.

Ans: $$v_O = \begin{cases} 5\ V, & v_I \le 0.6\ V, \quad Cutoff \\ 6.2 - 2v_I, & 0.6\ V < v_I \le 3\ V, \quad Active \\ 0.2\ V, & v_I > 3\ V, \quad Saturation \end{cases}$$



# Problem 9

- For the following circuit, $V_S = 10\ V$, $R_E = 100\ k\Omega$, and $R_I = 10\ k\Omega$. Assume $\beta = 100$ for the transistor, $V_{BE,\,active} = V_{BE,\,sat} = 0.6\ V$, and $V_{CE,\,at\ edge\ of\ sat} = V_{CE,\,sat} = 0.2\ V$.

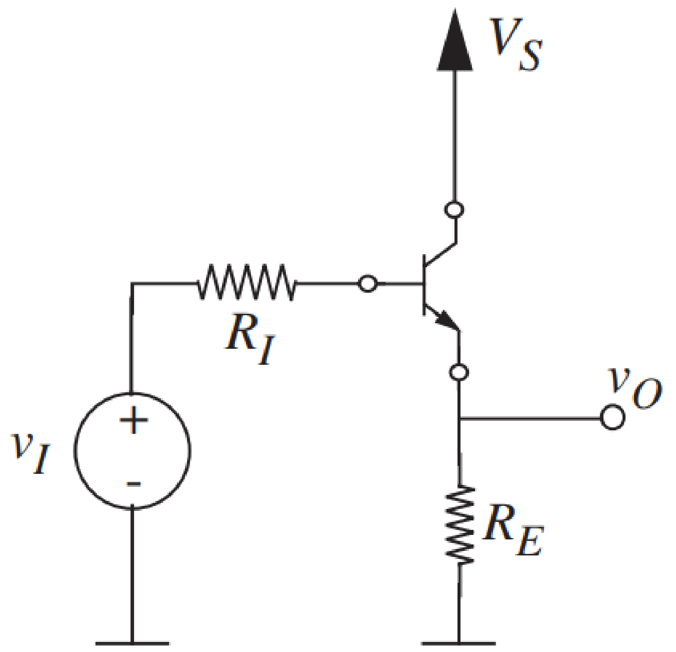
I. Derive an expression relating $v_O$ and $v_I$ assuming the transistor is in active mode. $\boxed{\text{Ans: } \boldsymbol{v_0 \approx v_I - 0.6}}$

II. What is the highest value of $v_I$ for which the transistor operates in cut-off. $\boxed{\text{Ans: } \boldsymbol{0.6\ V}}$

III. What are the highest (or lowest) values of $v_I$ for which the transistor remains in its active region (or operates at the edge of saturation). What are the corresponding range for $v_O$? *[$\beta = 100$ at the edge of saturation]* $\boxed{\text{Ans: } \boldsymbol{10.4\ V}}$

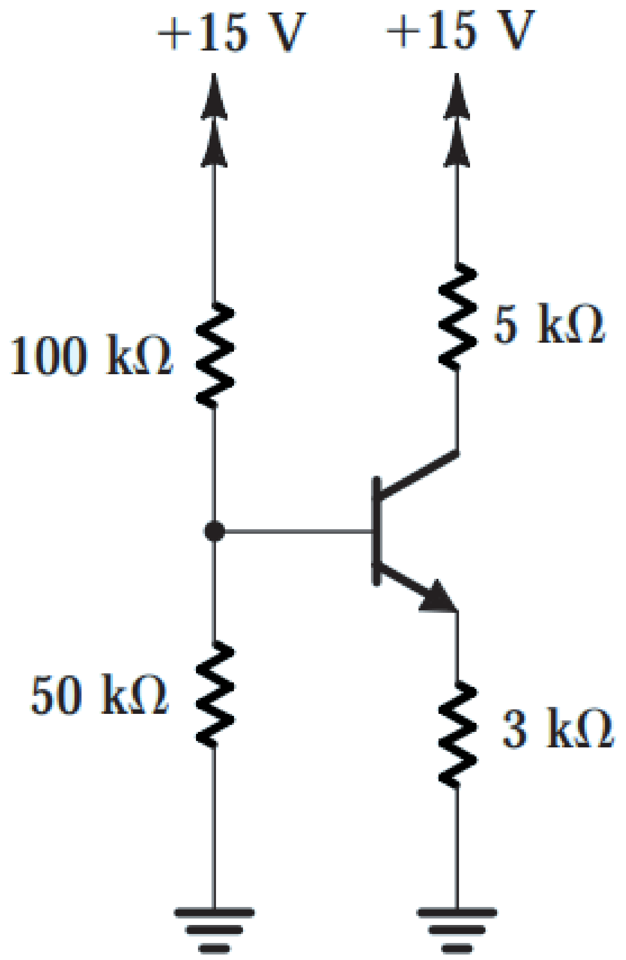
IV. Based on the expression found in I and values of $v_I$ and $v_O$ found in II and III, draw the VTC ($v_O\ vs.\ v_I$) graph for the circuit.

$$\text{Ans: } \boldsymbol{v_0} = \begin{cases} \boldsymbol{0\ V}, & \boldsymbol{v_I \le 0.6\ V}, \quad \boldsymbol{Cutoff} \\ \boldsymbol{v_I - 0.6}, & \boldsymbol{0.6\ V < v_I \le 10.4\ V}, \quad \boldsymbol{Active} \\ \boldsymbol{9.8\ V}, & \boldsymbol{v_I > 10.4\ V}, \quad \boldsymbol{Saturation} \end{cases}$$



# Problem 10

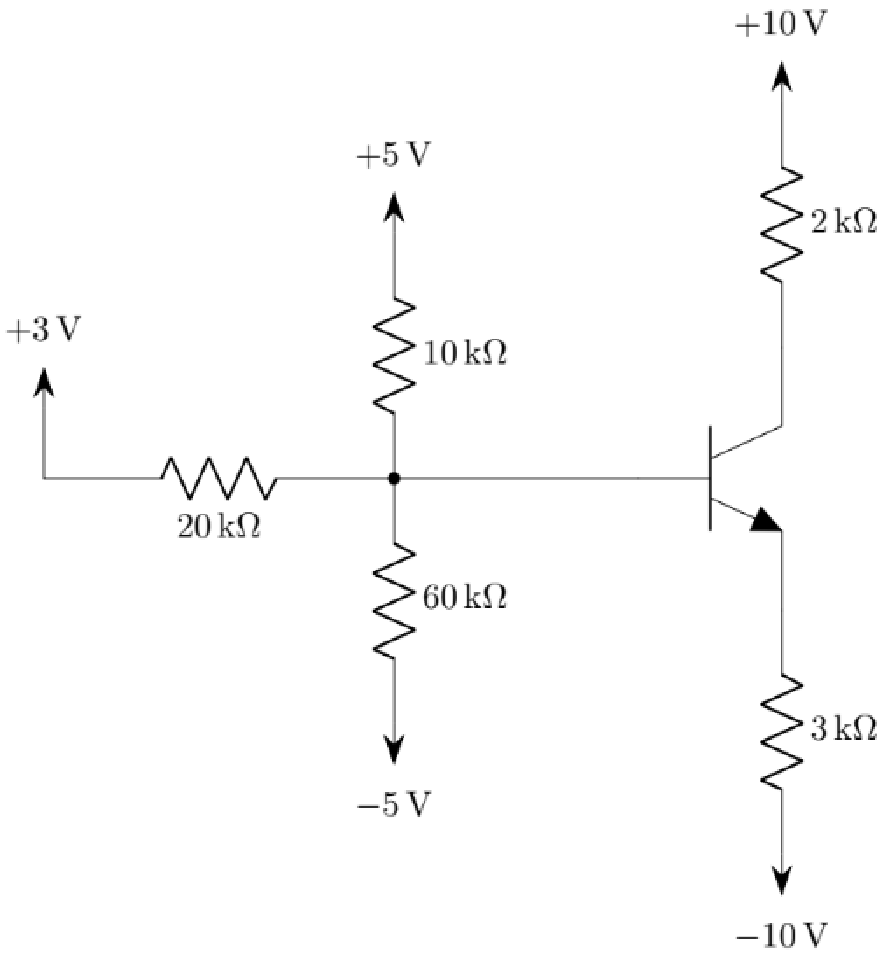
- For the NPN transistor in the following circuit, $\beta = 100$, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.8\,V$, and $V_{CE,\,sat} = 0.2\,V$. Determine the current through the $50\,k\Omega$ resistor.




Ans: $\boldsymbol{Active}$; $\boldsymbol{I_C = 1.28\,mA; I_E = 1.2928\,mA; I_B = 12.8\,\mu A; V_B = 4.58\,V;\ V_C = 8.6\,V; V_E = 3.88\,V; Current\ through\ 50\,k\Omega = 91.6\,\mu A}$

# Problem 11

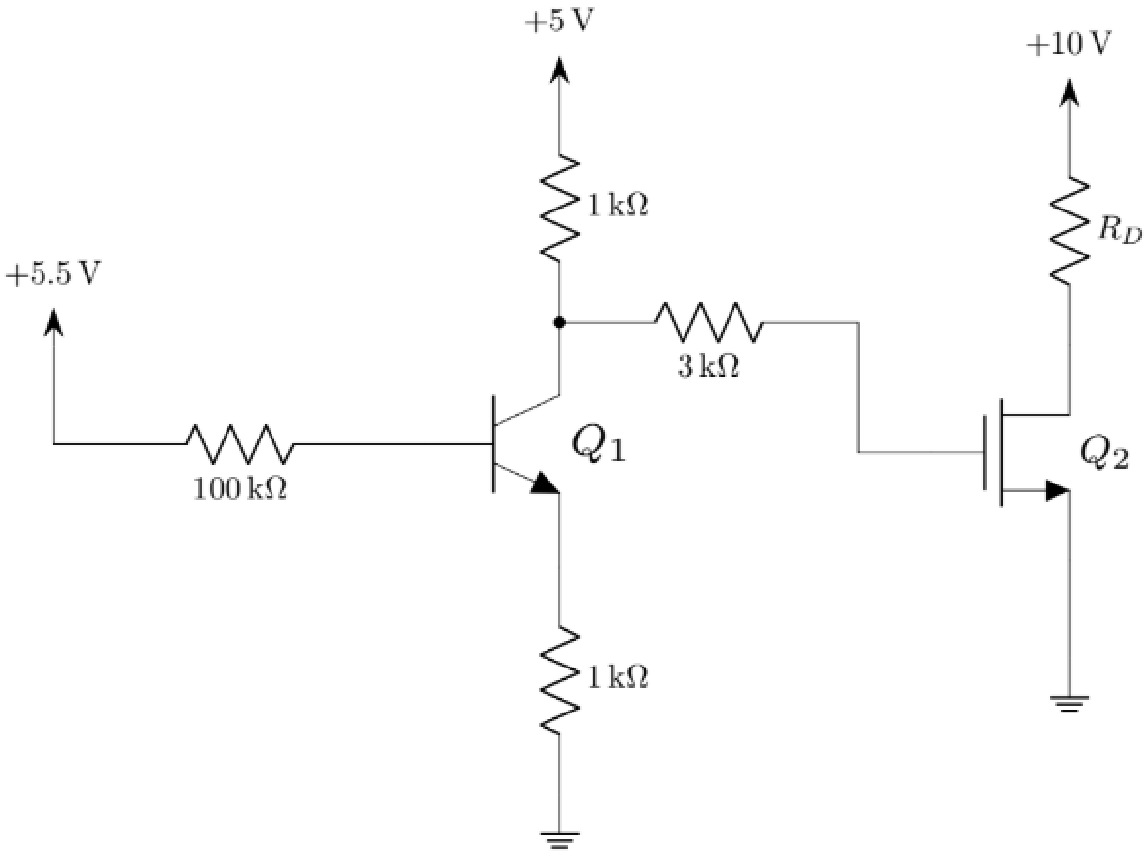
- For the NPN BJT in the adjacent circuit, $\beta = 100$ , $V_{BE,\,active} = 0.7\ V$ , $V_{BE,\,sat} = 0.8\ V$ , and $V_{CE,\,sat} = 0.2\ V$ . Determine the operating region of the transistor.




Ans: $\boldsymbol{Sat}$; $\boldsymbol{I_C = 3.9\ mA; I_E = 4\ mA; I_B = 0.1\ mA; V_B = 2.8\ V;\ V_C = 2.2\ V; V_E = 2\ V}$

# Problem 12

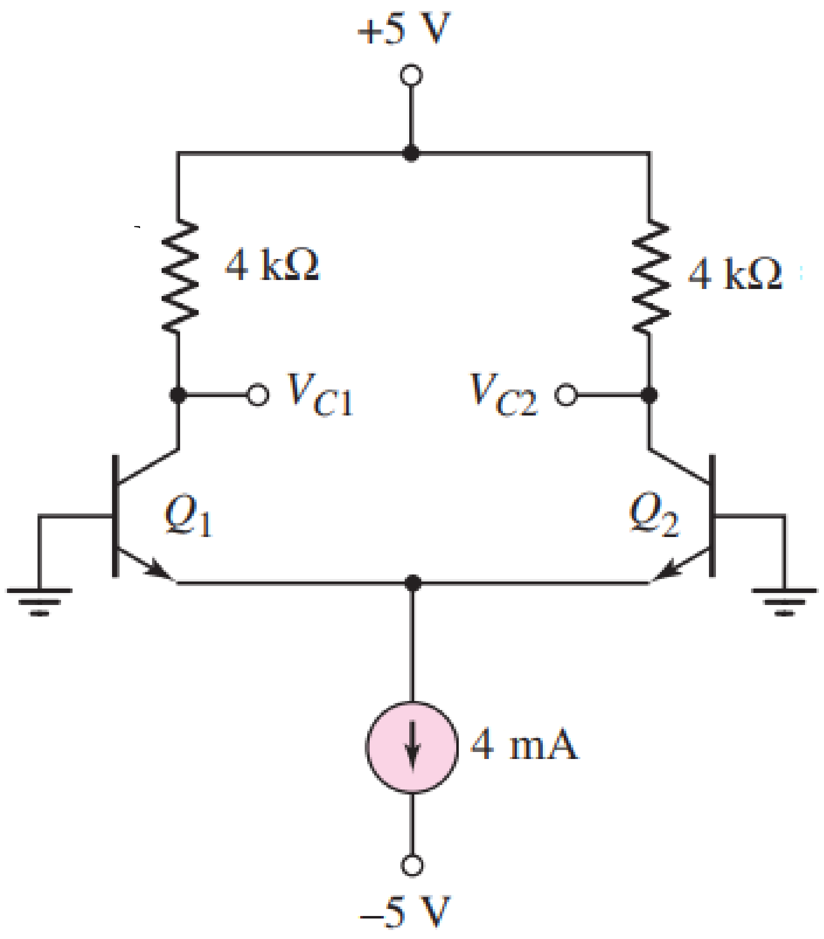
- For the transistor $Q_1$ in the adjacent circuit, $\beta = 100$, $V_{BE}$ is constant at $0.7\ V$ when $Q_1$ remains ON, and $V_{CE,\,sat} = 0.2\ V$. For $Q_2$, $k_n' = 0.25\ {}^{mA}/_{V^2}$, $\left(\frac{W}{L}\right) = \frac{0.72\ \mu m}{0.18\ \mu m}$, and $V_{Tn} = 1\ V$. Determine the value of $R_D$ so that $Q_2$ operates at the edge of saturation.




| | |
|---|---|
| Ans: $\boldsymbol{Q_1}$: $\boldsymbol{Active}$; $\boldsymbol{I_C = 2.39\ mA; I_E = 2.4139\ mA; I_B = 23.9\ \mu A; V_B = 3.11\ V;\ V_C = 2.61\ V; V_E = 2.4139\ V}$ | Ans: $\boldsymbol{Q_2}$: $\boldsymbol{V_G = 2.61\ V; V_D = 1.61\ V;\ I_{DS} = 1.2961\ mA;\ R_D \approx 7\ k\Omega}$ |

# Problem 13

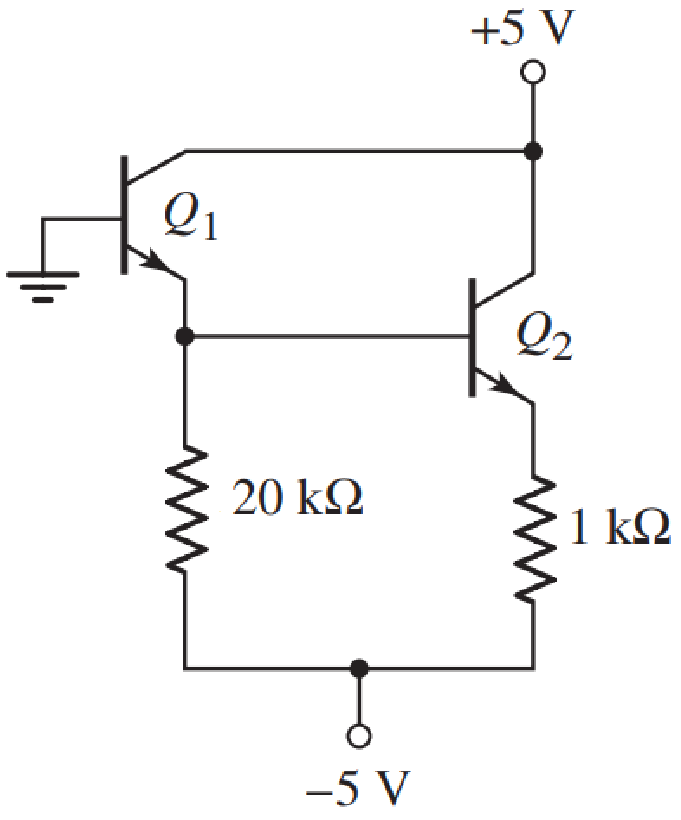
- The transistors in the adjacent circuit are identical. Here, $\beta = 100$, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.8\,V$, and $V_{CE,\,sat} = 0.2\,V$. Determine $V_{C1}$ and $V_{C2}$.




Ans: $\boldsymbol{Both\ Q_1\ \&\ Q_2\ are\ in\ Sat\ Mode;\ I_{E_1} = I_{E_2} = 2\ mA; V_{B_1} = V_{B_2} = 0\ V; V_{E_1} = V_{E_2} = -0.8\ V; V_{C_1} = V_{C_2} = -0.6\ V;\ I_{C_1} = I_{C_2} = 1.4\ mA; I_{B_1} = I_{B_2} = 0.6\ mA}$

# Problem 14

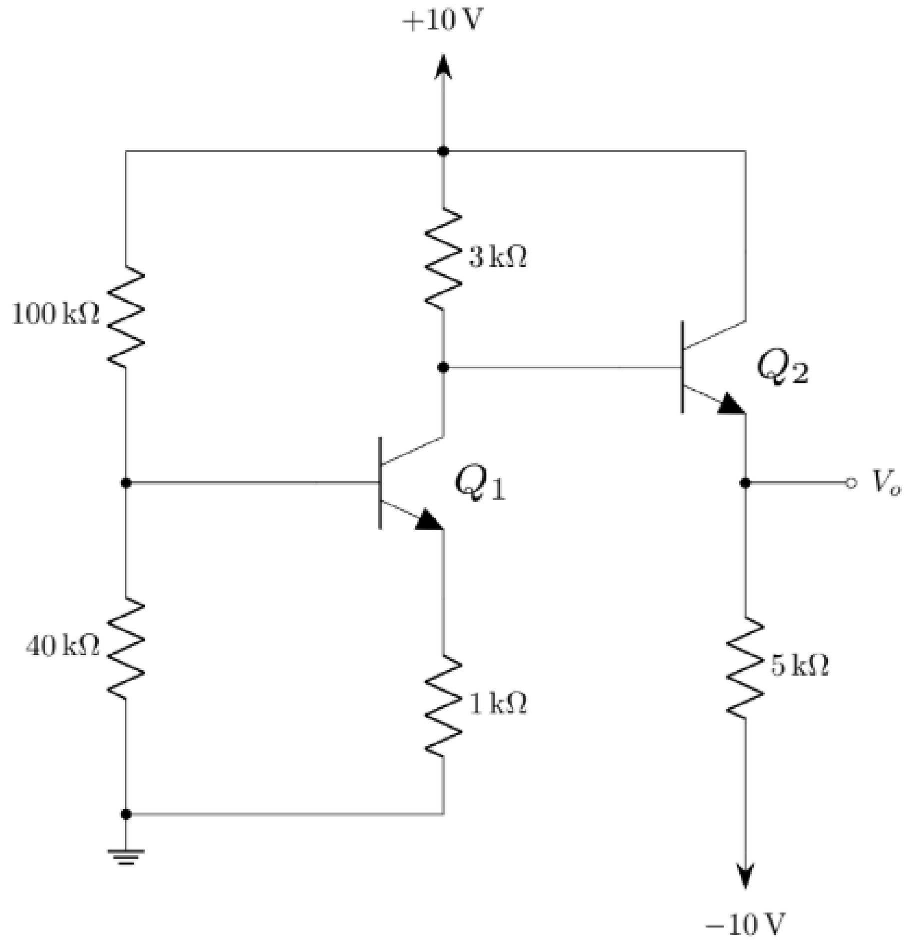
- The transistors in the following circuit are identical. Here, $\beta = 80$, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.7\,V$, and $V_{CE,\,sat} = 0.2$. Determine the base currents.




Ans: $\boldsymbol{Both\ Q_1\ \&\ Q_2\ are\ in\ Active\ Mode;\ V_{B_1} = V_{B_2} = 0\,V; V_{E_1} = V_{E_2} = -0.7\,V; I_{E_1} = I_{E_2} = 0.215\,mA; I_{B_1} = I_{B_2} = 2.7\,\mu A;\ V_{C_1} = V_{C_2} = 5\,V;\ I_{C_1} = I_{C_2} = 0.212\,mA}$

# Problem 15

- The transistors in the adjacent circuit are identical. Here, $\beta = 120$, $V_{BE,\,active} = 0.7\ V$, $V_{BE,\,sat} = 0.7\ V$ , and $V_{CE,\,sat} = 0.2\ V$ . Determine $V_o$.




Ans: $Q_1$ *is in Active Mode*; $I_{B_1} = 0.0238\ mA; I_{C_1} = 2.856\ mA; I_{E_1} = 2.88\ mA; V_{E_1} = 2.88\ V;\ V_{B_1} = 3.58\ V;\ V_{C_1} = V_{B_2} = 1.41\ V;$
$Q_2$ *is in Active Mode*: $V_{C_2} = 10\ V; I_{B_2} = 0.0177\ mA; I_{C_2} = 2.124\ mA; I_{E_2} = 2.1417\ mA; V_{E_2} = V_0 = 0.71\ V; V_{B_2} = V_{C_1} = 1.41\ V$

Up Next

# Relevant Questions from Past Semesters

# Final – Fall 2025 - 1/2

- The transistors $Q_2$ and $Q_3$ in the following circuit are identical and have the parameters given in the box.
  a. Determine $V_1$ and $V_2$.
  b. Determine $V_3$ and $I_{DS}$.

**NPN BJT Parameters:**

For the transistor $Q_2$,

Common emitter current gain: $\beta = 100$

Base-emitter voltage in the active region: $V_{BE(active)} = 0.7\,\text{V}$

Base-emitter voltage in the saturation region: $V_{BE(sat)} = 0.8\,\text{V}$

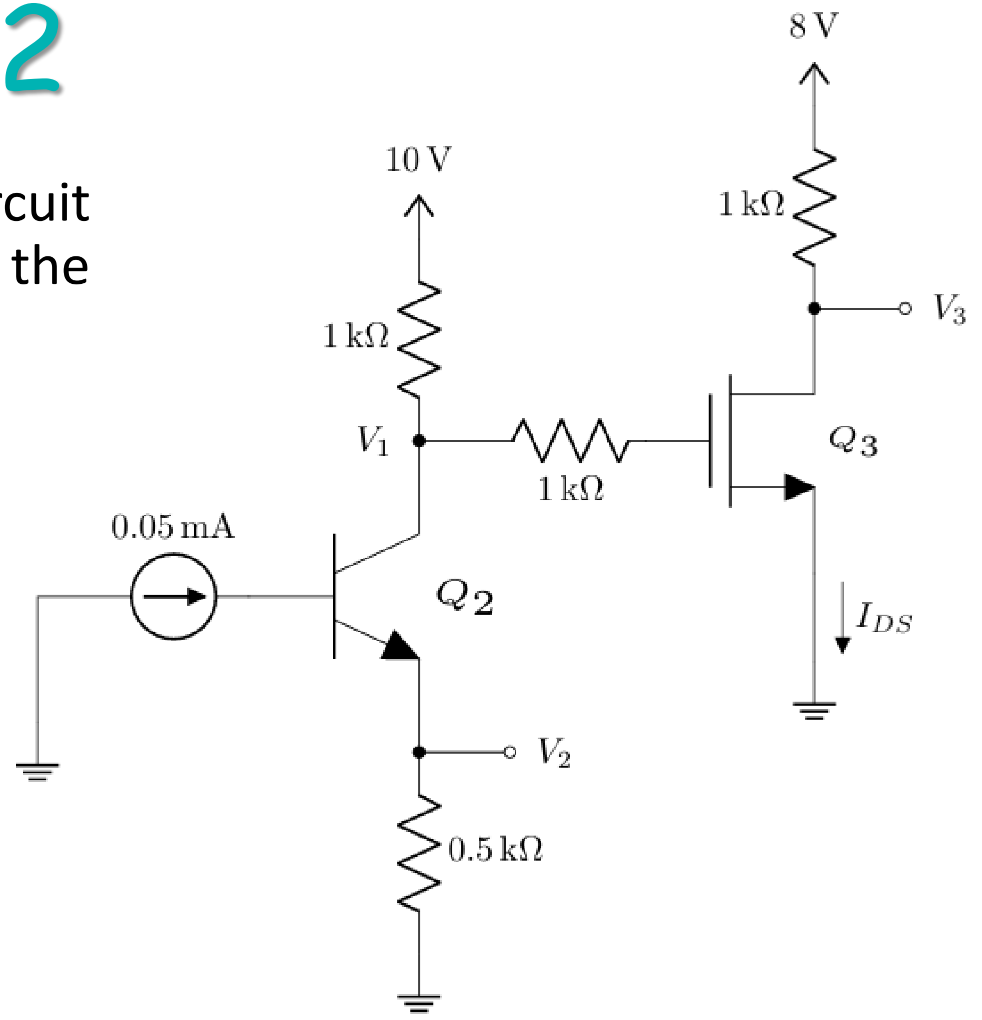
Collector-emitter voltage in the saturation region: $V_{CE(sat)} = 0.2\,\text{V}$

**N-Channel MOSFET Parameters:**

For the transistor $Q_3$,

Threshold voltage: $V_{T_n} = 1\,\text{V}$

Transconductance parameter: $k_n = 2\,\text{mA/V}^2$



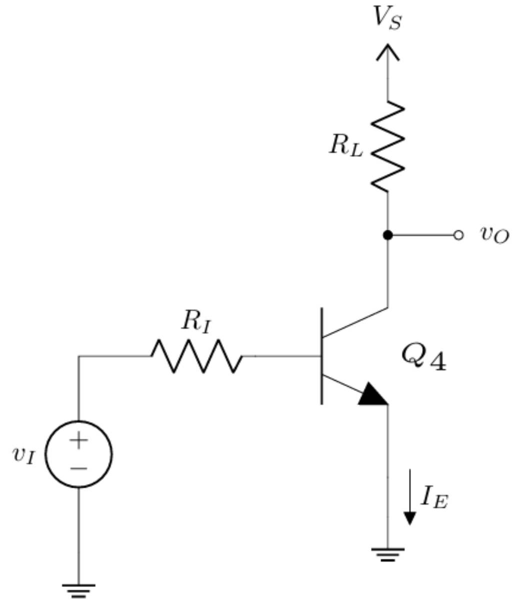

Ans: $(a)\ Q_2 \rightarrow Active;\ V_1 = 5\ V; V_2 = 2.525\ V$
$(b)\ Q_3 \rightarrow Triode; V_3 = 1\ V; I_{DS} = 7\ mA$

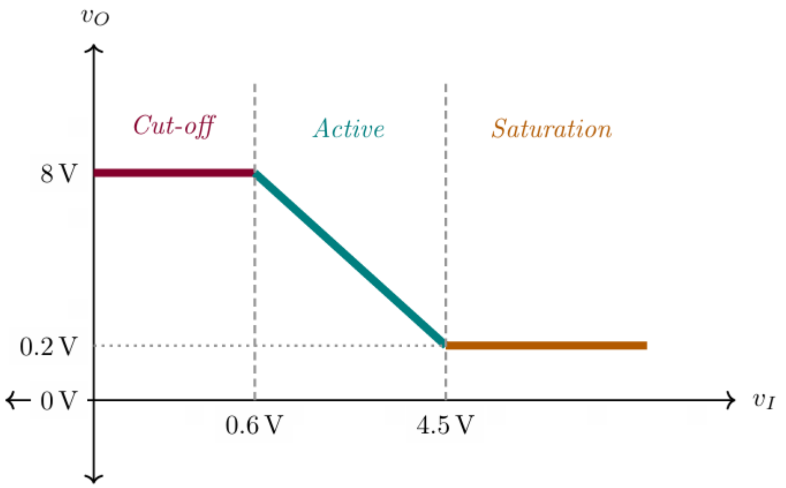
# Final — Fall 2025 - 2/2

The output voltage $v_O$ for the circuit shown below is plotted as a piecewise function of the input voltage $v_I$. The different regions of transistor operation are indicated by vertical dashed lines, and the specific transistor parameters are provided in the accompanying box.

a. When the transistor operates at the edge of saturation, the emitter current is $I_E = 3.94\ mA$. Determine the values of $R_L$ and $R_I$. [Use $\beta = 100$]

b. Using the values found in part $(a)$, what is the value of $I_E$ when $v_I = 5\ V$.




**NPN BJT Parameters:**

For the transistor $Q_4$,

Common emitter current gain: $\beta = 100$

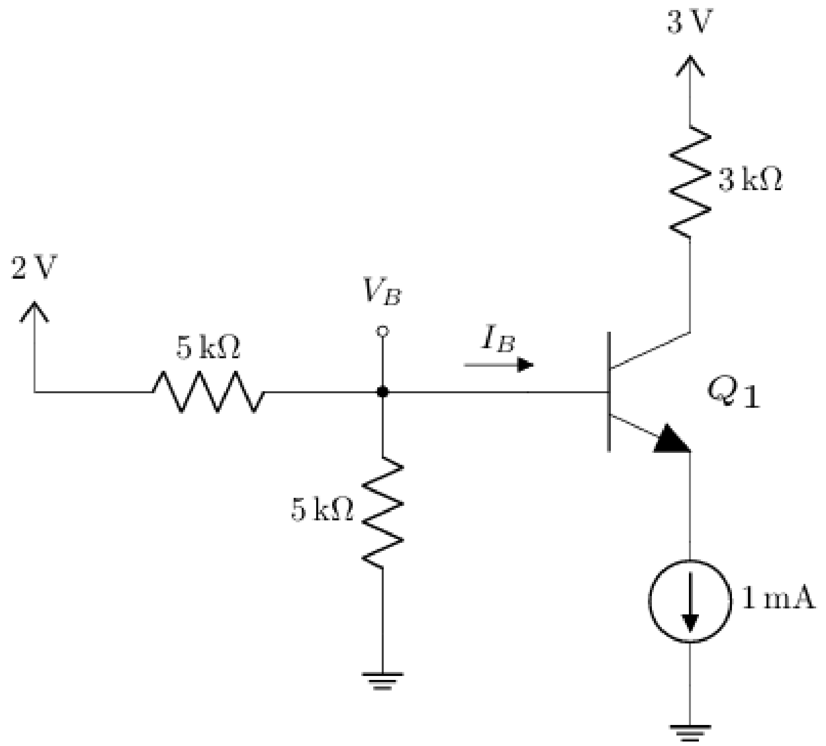
Base-emitter voltage: $V_{BE(sat)} = V_{BE(active)} \approx 0.6\,\text{V}$

Collector-emitter voltage: $V_{CE(sat)} = V_{CE(edge\ of\ sat)} \approx 0.2\,\text{V}$

Ans: $(a)\ \boldsymbol{R_I = 100\ k\Omega;\ R_L = 2\ k\Omega}$
$(b)\ \boldsymbol{I_E = 3.944\ mA;\ I_B = 0.044\ mA;\ I_C = 3.9\ mA}$

# Final — Summer 2025

- Verify that $Q_1$ operates in the saturation region, and determine all terminal currents and voltages of the transistor.




**NPN BJT Parameters:**

For the transistor $Q_1$:

Current gain: $\beta = 100$

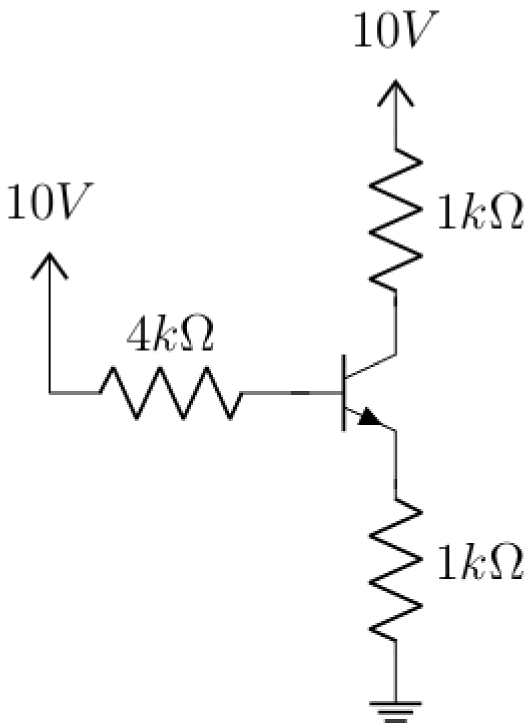
Base-emitter voltage in saturation: $V_{BE(sat)} = 0.8\,\text{V}$

Collector-emitter voltage in saturation: $V_{CE(sat)} = 0.2\,\text{V}$

Ans: $\boldsymbol{Sat}$; $\boldsymbol{I_C = 0.927\ mA}$; $\boldsymbol{I_E = 1\ mA}$; $\boldsymbol{I_B = 0.073\ mA}$; $\boldsymbol{V_B = 0.818\ V}$; $\boldsymbol{V_C = 0.218\ V}$; $\boldsymbol{V_E = 0.018\ V}$

# Final — Spring 2025

- Analyze the circuit below, and calculate $I_B$, $I_C$, $I_E$, $V_C$, and $V_E$ using the method of assumed states. You must validate your assumption.




**NPN BJT Parameters:**

For the transistor $Q_1$:

Current gain: $\beta = 100$

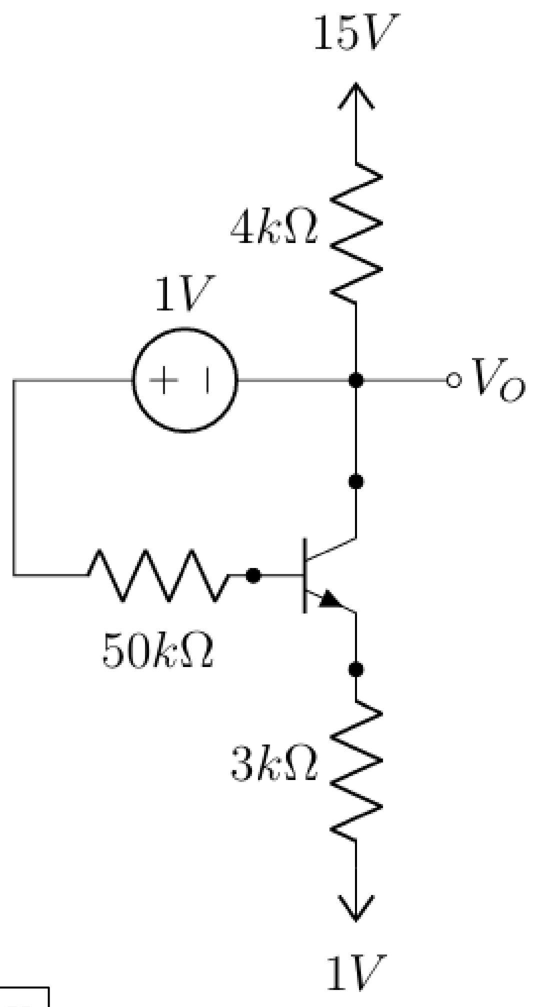
Base-emitter voltage in saturation: $V_{BE(sat)} = 0.8\,\text{V}$

Collector-emitter voltage in saturation: $V_{CE(sat)} = 0.2\,\text{V}$

Ans: $\boldsymbol{Sat};\ \boldsymbol{I_C = 4.42\ mA; I_E = 5.38\ mA; I_B = 0.96\ mA; V_B = 6.18\ V;\ V_C = 5.58\ V; V_E = 5.38\ V}$

# Final — Fall 2024

- Analyze the circuit below, and calculate $I_B$, $I_C$, $I_E$, and $V_O$ using the method of assumed states. You must validate your assumption.




**NPN BJT Parameters:**

For the transistor $Q_1$:

Current gain: $\beta = 100$

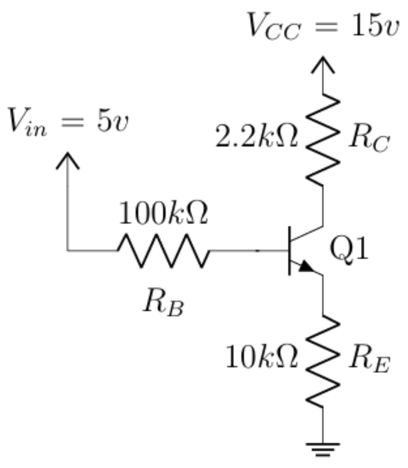
Base-emitter voltage in saturation: $V_{BE(sat)} = 0.8\,\text{V}$

Collector-emitter voltage in saturation: $V_{CE(sat)} = 0.2\,\text{V}$

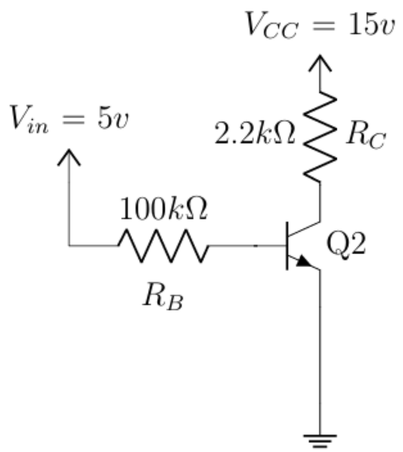
Ans: $\boldsymbol{Active}$; $\boldsymbol{I_C = 1.89\ mA; I_E = 1.908\ mA; I_B = 0.0189\ mA; V_B = 5.515\ V;\ V_C = 5.46\ V; V_E = 4.715\ V}$

# Final — Summer 2024 - 1/2

- Karim designed a BJT circuit. When his friend Rahim saw it, he claimed that the BJT would operate in the Active mode. Then Rahim built a circuit and asked Karim to figure out the operating mode the BJT.
  a. Calculate $I_B$, $I_C$, $I_E$, and $V_{CE}$ in Karim's circuit and determine whether Rahim was correct or not.
  b. Analyze Rahim's circuit and calculate $I_B$, $I_C$, $I_E$, and $V_{CE}$ to help Karim determine the operating state of the BJT.




***Karim's Circuit***




***Rahim's Circuit***

<u>For Both BJTs</u>

$\beta = 100$

$V_{BE(Active)} = 0.7v$

$V_{BE(Saturation)} = 0.8v$

$V_{CE(Saturation)} = 0.2v$

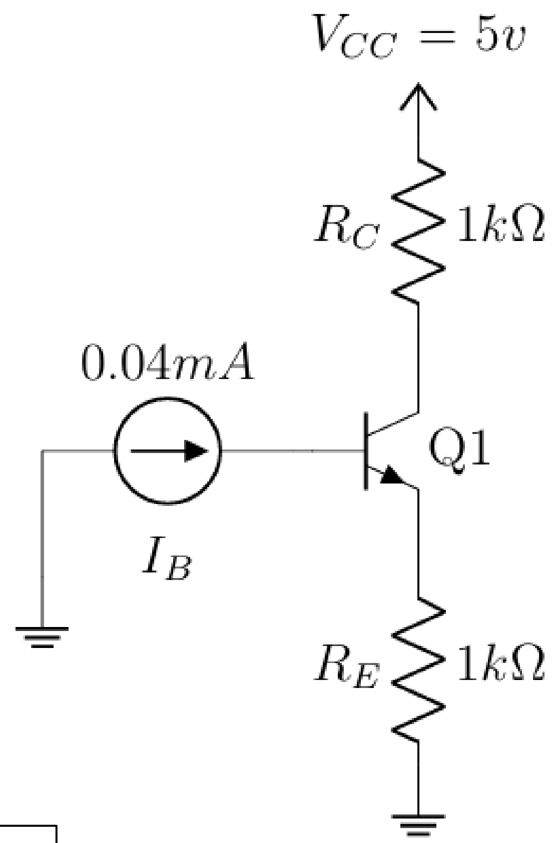
<u>Ans</u>: $(a)$ $\boldsymbol{Active;\ I_C = 0.3874\ mA; I_E = 0.391\ mA; I_B = 3.874\ \mu A; V_B = 4.61\ V;\ V_C = 14.15\ V; V_E = 3.91\ V}$
$(b)$ $\boldsymbol{Active;\ I_C = 4.3\ mA; I_E = 4.343\ mA; I_B = 0.043\ mA; V_B = 4.61\ V;\ V_C = 5.54\ V; V_E = 0\ V}$

# Final — Summer 2024 - 2/2

- Analyze the circuit below, and calculate $I_B$, $I_C$, $I_E$, and $V_{CE}$ using the method of assumed states. You must validate your assumption.

**BJT Parameters**

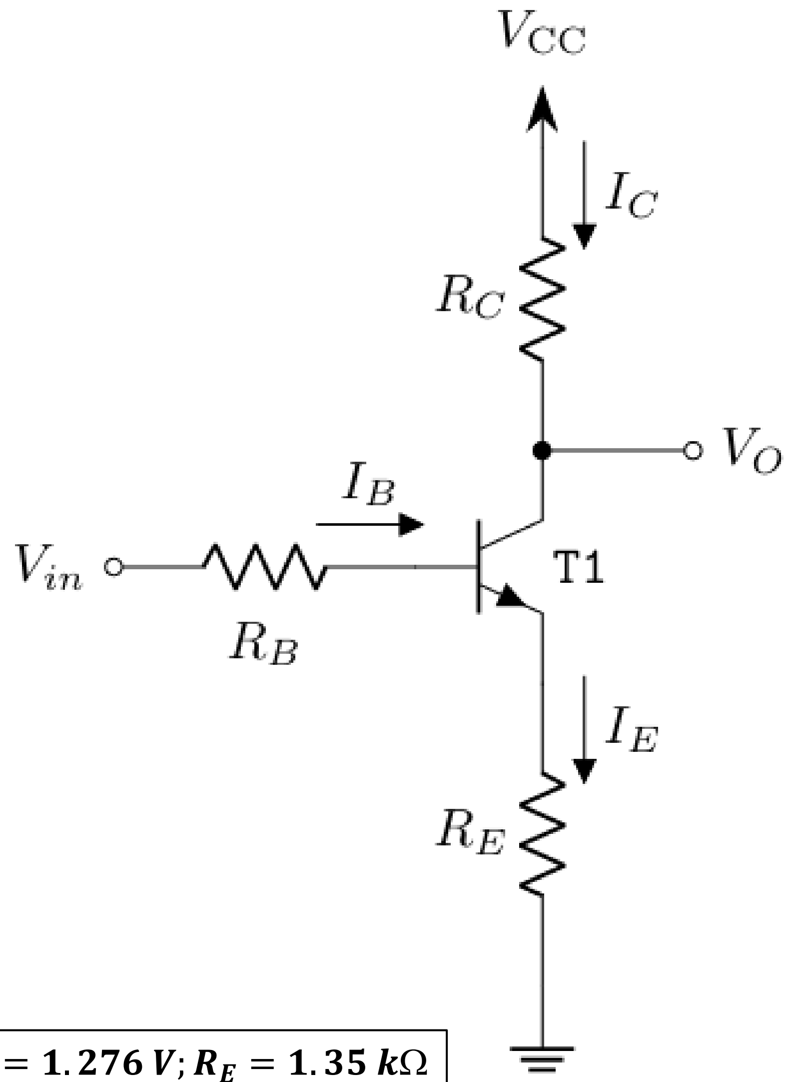
$$\beta = 90$$
$$V_{BE(Active)} = 0.7v$$
$$V_{BE(Saturation)} = 0.8v$$
$$V_{CE(Saturation)} = 0.2v$$




Ans: $\boldsymbol{Sat}$; $\boldsymbol{I_C = 4.93\ mA; I_E = 4.9825\ mA; I_B = 0.0525\ mA; V_B = 5.7825\ V;\ V_C = 5.46\ V; V_E = 4.715\ V}$

# Final — Spring 2024 - 1/2

- Andrew and Nicole found the adjacent circuit built in a trainer board. From the transistor model, they knew it had a gain of $\beta = 80$. Additionally, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.7\,V$, and $V_{CE,\,sat} = 0.2\,V$. They also saw $V_{CC} = 5\,V$, $R_B = 2\,k\Omega$, and $R_C = 3\,k\Omega$. However, the $R_E$ resistor was an unknown one. So, they provided an input of $V_{in} = 2\,V$ and measured the output to be $V_O = 2.2\,V$. Nicole said, "In this condition, the transistor is in active mode". But, Andrew disagreed.
    a. Design the circuit, i.e., determine the value of $R_E$, using what Nicole said about the mode of the transistor.
    b. Using the value of $R_E$ obtained from $(a)$, determine who will be right if $V_{in} = 4\,V$.




Ans: $(\boldsymbol{a})\ \boldsymbol{I_C = 0.933\,mA; I_E = 0.945\,mA; I_B = 0.012\,mA; V_B = 1.976\,V;\ V_C = V_O = 2.2\,V; V_E = 1.276\,V; R_E = 1.35\,k\Omega}$
$(\boldsymbol{b})\ \boldsymbol{Sat;\ I_C = 0.912\,mA; I_E = 1.53\,mA; I_B = 0.618\,mA; V_B = 2.766\,V;\ V_C = V_O = 2.264\,V; V_E = 2.066\,V;}$

# Final — Spring 2024 - 2/2

- For the adjacent circuit,
  a. Find out the currents going through the three terminals of the transistor Q1.
  b. Determine the the value of $V_1$ and $V_2$.
  c. Design the circuit, i.e. find an appropriate value of $R_D$, such that Q2 remains at the edge of saturation, i.e., when $V_{DS} = V_{GS} - V_T$.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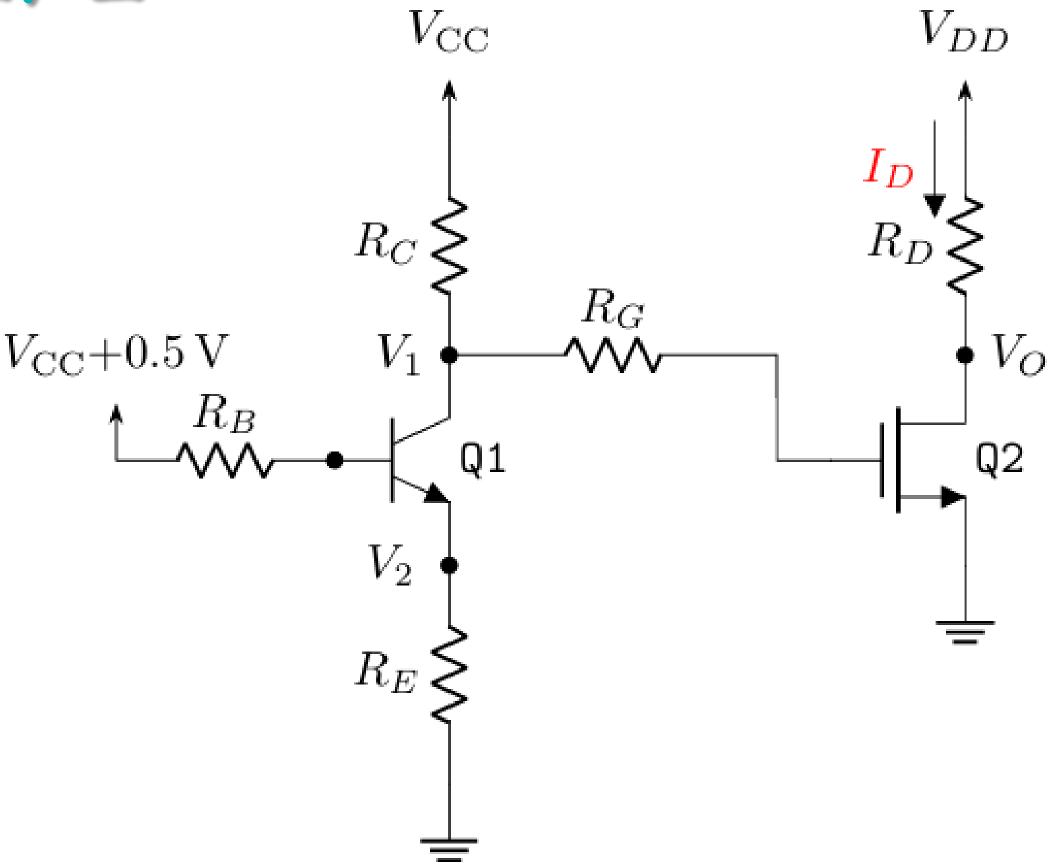


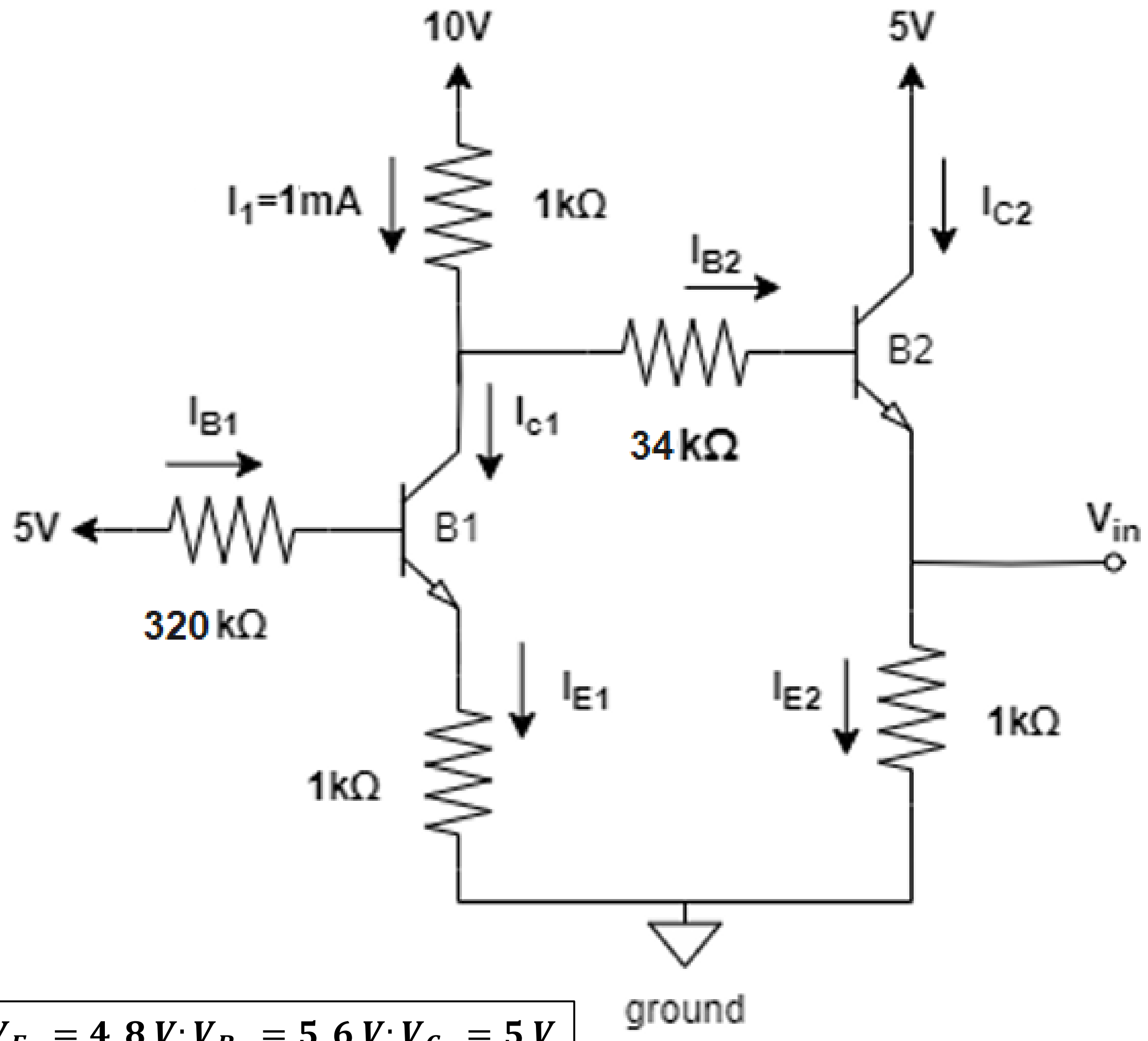

| Resistor values in (kΩ) | | | | |
|---|---|---|---|---|
| $R_B$ | $R_C$ | $R_E$ | $R_G$ | $R_D$ |
| 100 | 1.1 | 1 | 3 | - |

| Supply Voltages |
|---|
| $V_{CC}$ = 5 **V** |
| $V_{DD}$ = 10 **V** |

Ans: $(a)$ $\boldsymbol{Sat;}$ $\boldsymbol{I_C = 2.27\ mA; I_E = 2.295\ mA; I_B = 0.025\ mA; V_B = 3\ V}$
$(b)$ $\boldsymbol{V_1 = V_C = 2.503\ V; V_2 = V_E = 2.295\ V}$
$(c)$ $\boldsymbol{V_D = 1.5\ V; R_D = 7.556\ k\Omega}$

# Final – Fall 2023

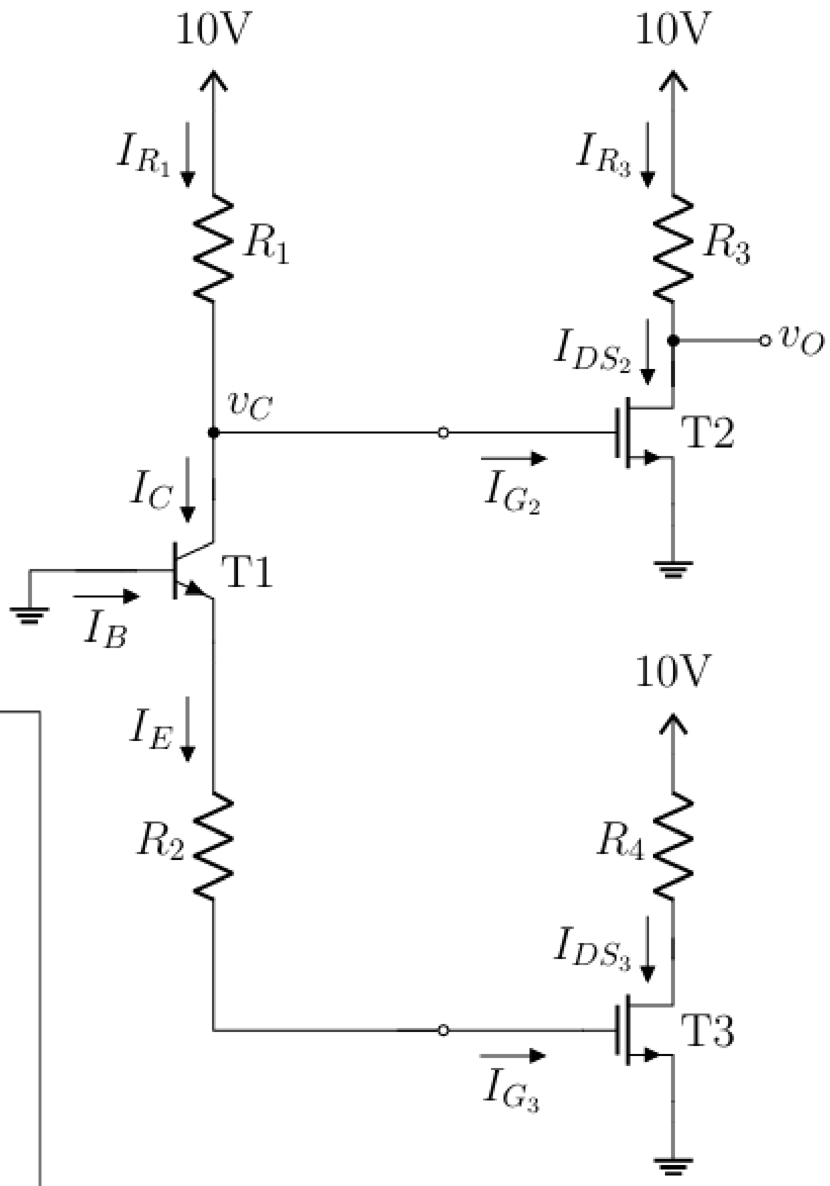
- The BJTs in the following circuit have parameters: $\beta = 75$, $V_{BE,\,active} = V_{BE,\,sat} = 0.8\,V$, and $V_{CE,\,sat} = 0.2\,V$. Determine the operating regions of $B_1$ and $B_2$.




Ans: $\boldsymbol{B2 \rightarrow Sat;\ I_{E_2} = 4.8\,mA; I_{C_2} = 4.7\,mA; I_{B_2} = 0.1\,mA; V_{E_2} = 4.8\,V; V_{B_2} = 5.6\,V; V_{C_2} = 5\,V}$
$\boldsymbol{B1 \rightarrow Active;\ I_{E_1} = 0.912\,mA; I_{C_1} = 0.9\,mA; I_{B_1} = 0.012\,mA; V_{E_1} = 0.912\,V; V_{B_1} = 1.16\,V; V_{C_1} = 9\,V}$

# Final — Summer 2023

- For the adjacent circuit, $R_1 = 1\ k\Omega$, $R_2 = 2\ k\Omega$, $R_3 = 3\ k\Omega$, and $R_4 = 4\ k\Omega$.
  a. Determine the values of $I_C$, $I_B$, $I_E$, and $v_C$ of T1.
  b. Determine the value of $I_{DS_2}$ and $v_O$ of T2.




**<u>for BJT</u>**
$\beta = 100$
$\alpha = 0.99$
$v_{BE(Active)} = 0.7V$
$v_{BE(Saturation)} = 0.8V$
$v_{CE(Saturation)} = 0.2V$

**<u>for MOSFET</u>**
$V_T = 1V$
$k = 5mA/V^2$

<u>Ans</u>: **T1** $\rightarrow$ $\boldsymbol{Cutoff}$; $\boldsymbol{I_E = I_B = I_C = 0\ mA; v_C = 10\ V; V_E > 0.8\ V; V_B = 0\ V}$;
**T3** $\rightarrow$ $\boldsymbol{Curoff}$; $\boldsymbol{I_{DS_2} = 0\ mA; I_{G_3} = 0\ mA}$
**T2** $\rightarrow$ $\boldsymbol{Triode}$; $\boldsymbol{I_{G_2} = 0\ mA; I_{DS_2} = 3.31\ mA; v_O = V_D = 0.074\ V}$

# Final – Spring 2023 - 1/3

- Analyze the following circuit to find the values of $i_B$, $i_C$, $i_E$, and $v_{CE}$. Assume saturation mode and validate the assumptions.

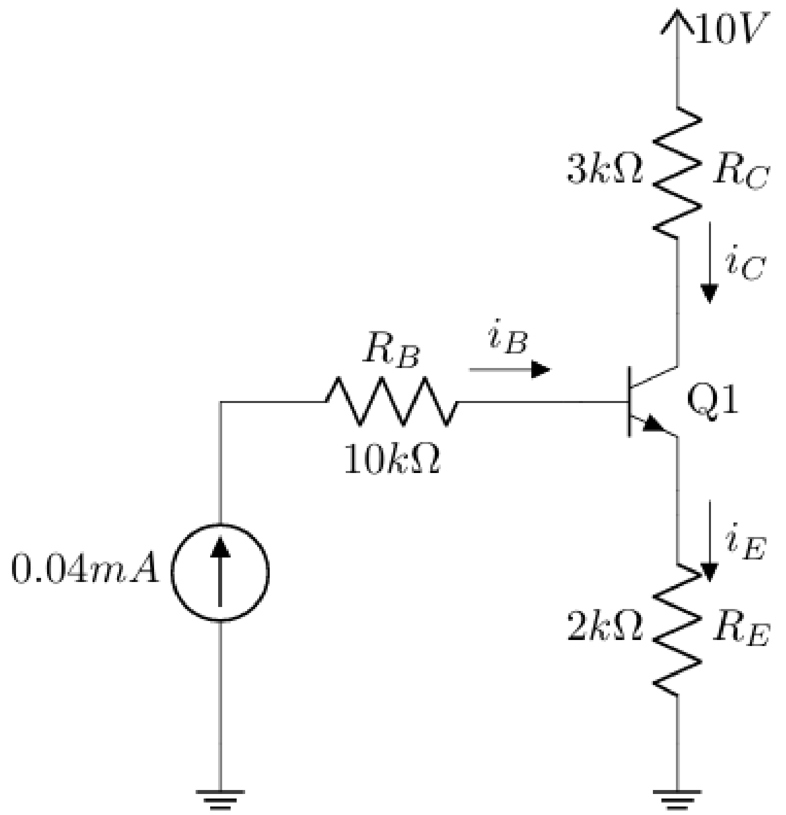
$\beta = 100$

$\alpha = 0.99$

$v_{BE(Active)} = 0.7V$

$v_{BE(Saturation)} = 0.8V$

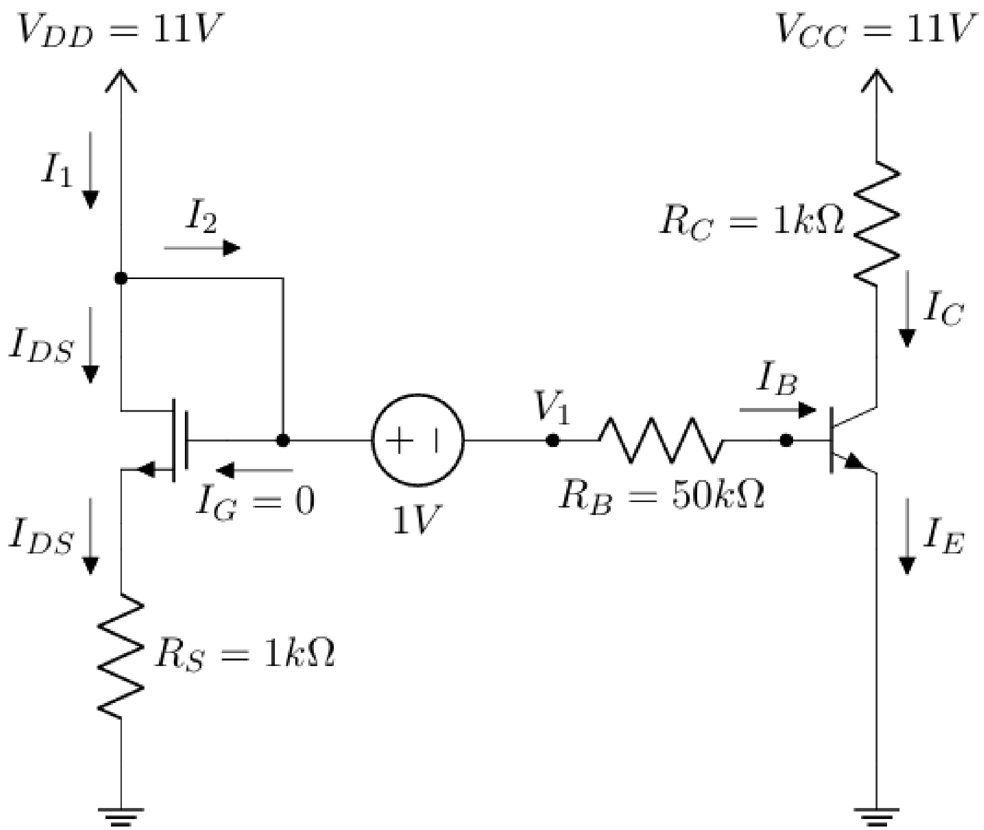
$v_{CE(Saturation)} = 0.2V$




Ans: $\boldsymbol{Sat}$; $\boldsymbol{i_C = 1.944\ mA}$; $\boldsymbol{i_E = 1.984\ mA}$; $\boldsymbol{i_B = 0.04\ mA}$; $\boldsymbol{V_B = 4.768\ V}$; $\boldsymbol{V_C = 4.168\ V}$; $\boldsymbol{V_E = 3.968\ V}$

# Final — Spring 2023 - 2/3

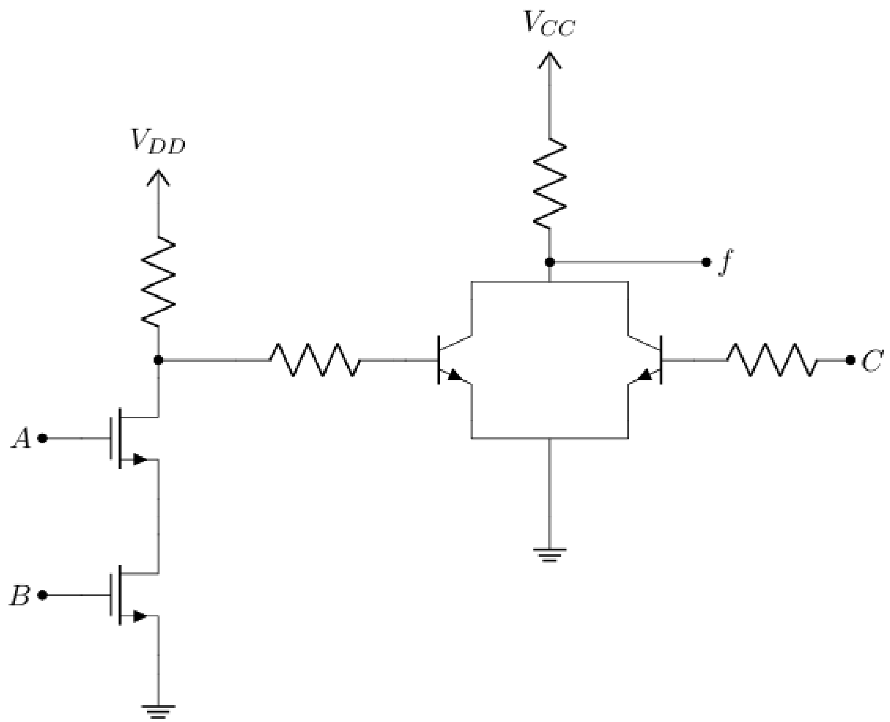
- The transistors in the following circuit have parameters: $k_n = 5\ {}^{mA}/_{V^2}$, $V_{T_n} = 0.7\ V$, $\beta = 85$, $V_{BE,\,active} = 0.7\ V$, $V_{BE,\,sat} = 0.8\ V$, and $V_{CE,\,sat} = 0.2\ V$.
   a. Find out the operating mode of the MOSFET. *[Hint: You don't need to solve the circuit to proof that.]*
   b. Calculate $I_{DS}$ and $V_{DS}$ using the given parameters.
   c. Determine $V_1$, $I_B$, $I_C$, and $I_E$.




Ans: $(b)$ $\boldsymbol{Sat;\ I_{DS} = 8.46\ mA; V_{DS} = 8.46\ V}$
$\boldsymbol{(c)\ Sat;\ V_1 = 10\ V; I_B = 0.184\ mA; I_C = 10.8\ mA; I_E = 10.984\ mA;\ V_C = 0.2\ V; V_B = 0.8\ V; V_E = 0\ V}$

# Final — Spring 2023 - 3/3

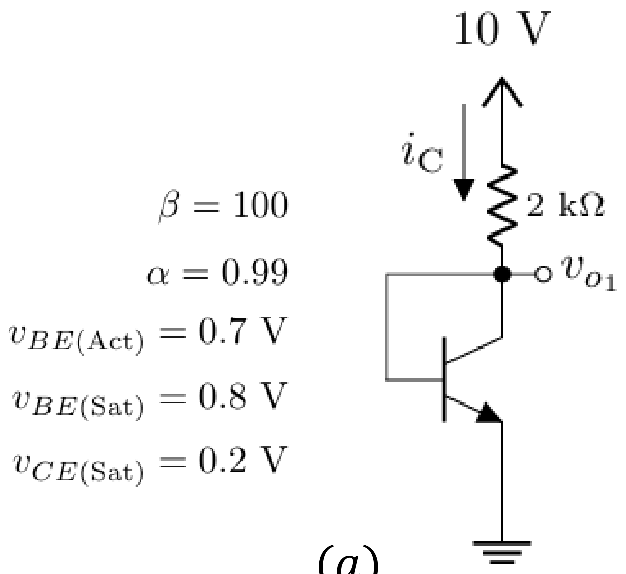
a. For the logic circuit $(b)$, write an expression for the Boolean output $f$ in terms of the Boolean inputs $A$, $B$, and $C$.



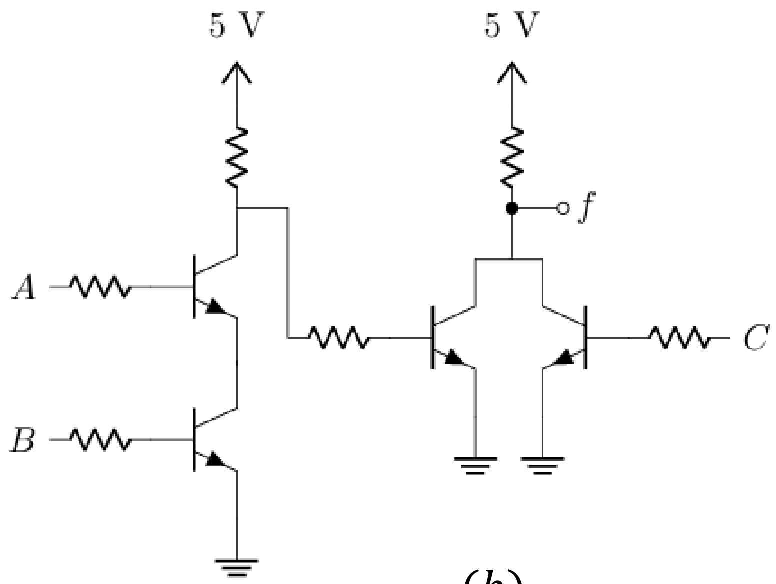
# Final – Fall 2022

a. Analyze the circuit $(a)$ to find $i_C$ and $v_{O1}$ using the Method of Assumed State. Validate your assumptions.

b. For the logic circuit $(b)$, write an expression for the Boolean output $f$ in terms of the Boolean inputs $A$, $B$, and $C$.




$(a)$

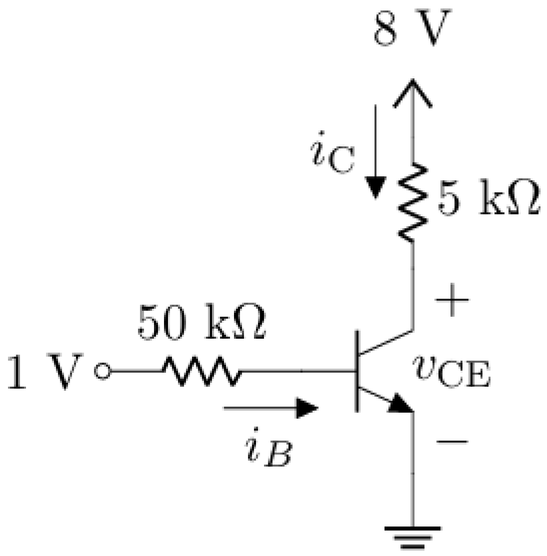



$(b)$

Ans: $(a)$ $\boldsymbol{Active}$; $\boldsymbol{i_C = 4.65\ mA; i_E = 4.6965\ mA; I_B = 0.0465\ mA; V_B = V_C = v_{o1} = 0.7\ V;\ V_E = 0\ V}$

# Final – Summer 2022

- Analyze the adjacent circuit to find $i_C$ and $v_{CE}$ using the Method of Assumed State. Validate your assumptions.




$\beta = 100$

$\alpha = 0.99$

$v_{BE(\text{Active})} = 0.7\text{ V}$

$v_{BE(\text{Saturation})} = 0.8\text{ V}$

$v_{CE(\text{Saturation})} = 0.2\text{ V}$

Ans: $\boldsymbol{Active;\ i_C = 0.6\,mA; i_E = 0.6006\,mA; I_B = 0.006\,mA; V_B = 0.7\,V;\ V_E = 0\,V; V_C = V_{CE} = 5\,V}$

# Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:


1. Agarwal, A. and Lang, J., Foundations of Analog and Digital Electronic Circuits, Morgan Kaufmann
2. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
3. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill